ST314 Group Project

Emancipative Values: A Multilevel Analysis

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1 Introduction

Emancipative values, defined by orientations emphasising personal autonomy, freedom of choice, and equality of opportunity, are known to be important indicators of modern society. Using data from Wave 7 of the World Values Survey (2017–2022), this project investigates how these values vary across individuals and national contexts. The outcome variable, *emaval*, ranges from 0 to 100, with higher values indicating stronger support for emancipative principles. The sample comprises approximately 76,500 individuals from 53 countries, forming a two-level hierarchical structure of individuals and countries.

We specifically address five research questions: quantifying country-level variance (RQ1), identifying individual predictors (RQ2), testing for gender-specific effects (RQ3), assessing contextual influences (RQ4), and evaluating macro-level drivers such as economic development and democracy (RQ5). Given the structure of the data, we used multilevel random-intercept modelling strategies, progressing from a baseline null model to models incorporating interactions, contextual and country-level predictors.

2 Descriptive Analysis

2.1 Univariate Statistics

2.1.1 Response Variable

The response variable `emaval` has a theoretical range of 0 to 100, with higher values indicating stronger emancipative orientations. The distribution is slightly right-skewed (mean = 43.11, median = 41.00), with the majority of scores between 20 and 60.

2.1.2 Individual-Level Predictors (Level 1)

Summary statistics are shown in Table 1. Age is roughly symmetrically distributed around the early forties. Education is right-skewed, with most respondents concentrated at lower-to-middle levels; although strictly ordinal, it is treated as continuous in the models. The sample is roughly evenly split by sex. A majority of respondents have a co-resident partner and live in urban areas. Most respondents are employed, with 33.1% economically inactive and 7.9% unemployed.

Table 1: Summary statistics for individual-level variables

Panel A: Continuous variables				
Variable	Mean	SD	Median	(Theoretical) Range
emaval	43.11	18.23	41.00	0–100
age	42.90	16.36	41.00	16–103
educ	3.47	2.03	3.00	0–8
Panel B: Categorical/Binary variables				
Variable	Category		Proportion (%)	
female	Female		52.4	
	Male		47.6	
partner	Has partner		64.9	
	No partner		35.1	
rural	Urban		64.4	
	Rural		35.6	
empstat	Employed		59.0	
	Economically inactive		33.1	
	Unemployed		7.9	

2.1.3 Country-Level Predictors (Level 2)

Country-level predictors show considerable variation across these countries as shown in Table 2. The democracy score (polity) ranges from −7.0 to 10.0 (mean = 5.49, SD = 5.33), showing that the sample includes both autocratic and democratic regimes. GDP per capita ranges widely from \$2,312 to \$65,281 (mean = \$21,267, SD = \$16,704), suggesting substantial differences in economic development across

the countries. Income inequality (incrichest10p) is less variable, ranging from 20.9% to 41.9% (mean = 28.88%, SD = 4.76).

Table 2: Summary statistics for country-level variables ($N = 53$ countries)

Variable	Mean	SD	Min	Max
polity	5.49	5.33	−7.0	10.0
GDPpercap1 (USD)	21,267	16,704	2,312	65,281
incrichest10p (%)	28.88	4.76	20.9	41.9

2.2 Between and Within-Country Variation

Please see Figure A1 for reference.

2.2.1 Between-Country Variation

We use country means to represent each country's overall level of emancipative values. The country means range from 23.63 (Egypt) to 67.75 (Germany), with a spread of over 44 points on a 100-point scale. This indicates a substantial between-country variation in emancipative values. Western European countries, including Germany, the Netherlands, and Great Britain cluster at the top of the distribution, while countries in North Africa and the Middle East sit at the lower end. The overall country mean is 43.11 (median = 40.67). We conclude that roughly half of the surveyed countries fall below the overall individual-level mean of 43.11, with a small number of high-scoring countries pulling the distribution upward.

2.2.2 Within-Country Variation

To illustrate within-country variation, we selected Egypt, Mexico, and Germany as mentioned above, representing the bottom, middle, and top of the country-mean distribution. Across all three countries, substantial within-country spread is evident, confirming that country-level means alone cannot fully characterise the distribution of `emaval`.

2.3 Bivariate Associations between Emancipative Values and Predictors

Please see Figure A2 for reference.

2.3.1 Associations with Individual-Level Predictors

Age shows almost no association with `emaval` ($r = 0.049$), with mean scores varying by fewer than 4 points across the full age range, suggesting a limited marginal effect at the individual level.

There are notable differences across the binary predictors. Women score slightly higher than men (mean = 44.26 vs. 41.85), and those without a co-resident partner score higher than those with one (mean = 45.46 vs. 41.84). The largest gap is for rural residence: urban respondents score nearly 9 points higher than rural respondents (mean = 46.27 vs. 37.40), suggesting area of residence is an important predictor.

Education shows the strongest individual-level association ($r = 0.324$), with mean `emaval` rising from 31.52 at level 0 to 57.14 at level 8. The increase is most pronounced at lower education levels (0–4), with a small dip at level 5, but the overall trend is largely monotonic.

2.3.2 Associations with Country-Level Predictors

At the country level, we find that GDP per capita has the strongest association with mean emaval (r = 0.817). Wealthier countries show considerably higher emaval values on average. The relationship appears broadly linear across the full range of GDP, though there is notable scatter at lower income levels. Democracy score (polity) also shows a moderate positive association (r = 0.461), with more democratic countries tending to have higher mean emaval, though the considerable spread at polity = 10 indicates that democratic governance alone does not fully determine a country's emancipative values. In contrast, income inequality (incrichest10p) shows weak linear association with mean emaval (r = -0.139), and the scatterplot confirms that the relationship is weak and diffuse across the full range of the variable.

3 Modelling and Results

3.1 RQ1: Do countries differ in `emaval`?

To assess the extent of country differences in emancipative values, we first used a variance components model:

$$y_{ij} = \beta + u_j + e_{ij},$$

where y_{ij} denotes the emancipative values score of individual i in country j, β is the overall mean, $u_j \sim N(0, \sigma_u^2)$ is the country-level random intercept, and $e_{ij} \sim N(0, \sigma_e^2)$ is the individual-level residual.

The estimated overall mean emancipative value was $\hat{\beta} = 41.93$. The estimated country-level variance was $\hat{\sigma}_u^2 = 118.6$, and the individual-level variance was $\hat{\sigma}_e^2 = 192.3$.

The intraclass correlation coefficient (ICC) was calculated as:

$$\text{ICC} = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \hat{\sigma}_e^2} = \frac{118.6}{118.6 + 192.3} = 0.381.$$

This indicates that approximately 38% of the total variance in emancipative values lies between countries, with the remaining 62% occurring within countries (or between individuals). These results attribute a substantial portion of variance to country-level differences and suggest that countries do indeed differ in emancipative values.

3.2 RQ2: How do individual characteristics relate to `emaval`?

To determine an answer, we included predictors at the individual level in the model:

$$\begin{aligned} y_{ij} = \beta_0 &+ \beta_1 \text{female}_{ij} + \beta_2 \text{age}_{ij} + \beta_3 \text{educ}_{ij} + \beta_4 \text{partner}_{ij} + \beta_5 \text{rural}_{ij} \\ &+ \beta_6 \text{Unemployed}_{ij} + \beta_7 \text{Inactive}_{ij} + u_j + e_{ij}, \end{aligned}$$

where notation follows RQ1. Unemployed and Inactive are dummy variables for `empstat`, and the baseline for `empstat` is `Employed`. Results are printed in Table A1 in the Appendix.

The likelihood ratio test that compared this model with the initial RQ1 model (without individual-level predictors) was highly significant, indicating that individual characteristics substantially improve model fit. As shown in Table A1, all individual-level predictors were statistically significant, with education showing the strongest positive association and rural residence the largest negative effect.

In terms of variance components, the estimated country-level variance decreased from $\hat{\sigma}_u^2 = 118.6$ in RQ1 to $\hat{\sigma}_u^2 = 107.1$ after including individual-level predictors. This corresponds to a proportional reduction of about 9.7% in between-country variance. The ICC declined only slightly, from 0.381 to 0.372), indicating that most between-country variance in emancipative values remains unexplained by differences in individuals alone.

3.3 RQ3: Do effects differ for men and women?

To determine an answer, we used interaction terms between `female` and each individual-level variable. Each interaction was first tested separately against the RQ2 model using likelihood ratio tests, which are shown in Table A2, followed by estimation of a full interaction model including all cross-products. The full model can be written as:

$$\begin{aligned} y_{ij} = \beta_0 &+ \beta_1 \text{female}_{ij} + \beta_2 \text{age}_{ij} + \beta_3 \text{educ}_{ij} + \beta_4 \text{partner}_{ij} + \beta_5 \text{rural}_{ij} \\ &+ \beta_6 \text{Unemployed}_{ij} + \beta_7 \text{Inactive}_{ij} \\ &+ \beta_8 (\text{female}_{ij} \times \text{age}_{ij}) + \beta_9 (\text{female}_{ij} \times \text{educ}_{ij}) \\ &+ \beta_{10} (\text{female}_{ij} \times \text{partner}_{ij}) + \beta_{11} (\text{female}_{ij} \times \text{rural}_{ij}) \\ &+ \beta_{12} (\text{female}_{ij} \times \text{Unemployed}_{ij}) + \beta_{13} (\text{female}_{ij} \times \text{Inactive}_{ij}) + u_j + e_{ij}, \end{aligned}$$

where men serve as the reference group.

All interactions except rural residence remained statistically significant in the full model (Table A3). The main effects represent associations for men. Age is negatively associated with emancipative values for men ($\hat{\beta} = -0.068$), with the interaction term ($\hat{\beta} = -0.031$) indicating a stronger negative effect for women. Education is positively associated for men ($\hat{\beta} = 1.072$), and the positive interaction ($\hat{\beta} = 0.319$)

suggests this effect is stronger for women. Partnership is negatively associated for men ($\hat{\beta} = -0.778$), with a substantially larger reduction for women ($\hat{\beta} = -0.864$). Similarly, the negative interactions for unemployment and inactivity indicate that labour market disadvantages are more strongly associated with lower emancipative values among women than men. The non-significant rural interaction suggests the urban–rural gap does not differ meaningfully by gender. Notaby, the rural interaction was significant when tested individually (Table A2, $p = 0.004$), but lost significance in the full model ($p = 0.203$), likely due to collinearity with the other interaction terms.

3.4 RQ4: Contextual (between vs within) effects

To determine an answer, we decomposed each Level-1 predictor x_{ij} into its individual component and the corresponding country mean $\bar{x}_j$. Each contextual term was first tested separately against the RQ2 model using likelihood ratio tests, which are shown in Table A4, followed by estimation of a full contextual model including all country means. The model takes the form:

$$\begin{aligned} y_{ij} = \beta_0 &+ \beta_{W1}\,\text{female}_{ij} + \beta_{W2}\,\text{age}_{ij} + \beta_{W3}\,\text{educ}_{ij} + \beta_{W4}\,\text{partner}_{ij} + \beta_{W5}\,\text{rural}_{ij} \\ &+ \beta_{B1}\,\overline{\text{female}}_j + \beta_{B2}\,\overline{\text{age}}_j + \beta_{B3}\,\overline{\text{educ}}_j + \beta_{B4}\,\overline{\text{partner}}_j + \beta_{B5}\,\overline{\text{rural}}_j \\ &+ \gamma_1\,\text{Unemployed}_{ij} + \gamma_2\,\text{Inactive}_{ij} + u_j + e_{ij}. \end{aligned}$$

The β_Ws capture within-country (or individual-level) effects, while the β_Bs capture between-country effects. Contextual effects exist when $\beta_B \neq \beta_W$. Results are shown in Table A5.

A likelihood ratio test comparing the full contextual model with the within-only model (identical to our RQ2 model) was highly significant ($\chi^2(5) = 48.61$, $p < 0.001$), indicating that country-level means jointly improve model fit.

By observing the individual LRTs in Table A4, significant contextual effects were found for age, partner, rural, and educ, while `female_cm` was not significant in isolation ($p = 0.211$). In the full contextual model (Table A5), age shows a clear contextual reversal: the within-country effect is negative ($\hat{\beta}_W = -0.082$), while the between-country effect is positive ($\hat{\beta}_B = 1.19$, $p < 0.001$), indicating that countries with higher average age exhibit higher overall emancipative values despite older individuals scoring lower. For partner status, both effects are negative ($\hat{\beta}_W = -1.203$; $\hat{\beta}_B = -26.504$, $p = 0.024$). Education and rural composition show no significant contextual effects ($p = 0.505$; $p = 0.149$). It is worth noting that `female_cm` became significant in the full model ($p = 0.012$), suggesting its effect is contingent on controlling for the other country-mean predictors.

3.5 RQ5: Country characteristics (controlling for individual and contextual effects)

We began from the contextual specification in RQ4 (individual-level predictors plus their country means) and then added country-level characteristics: a democracy score (`polity`), GDP per capita (rescaled as `GDP_10k` = `GDPpercap1`/10,000), and income share of the richest 10% (`incrichest10p`). Let i index individuals and j index countries. Our final model can be written as:

$$\begin{aligned} y_{ij} = \beta_0 &+ \beta_1\text{female}_{ij} + \beta_2\text{age}_{ij} + \beta_3\text{educ}_{ij} + \beta_4\text{partner}_{ij} + \beta_5\text{rural}_{ij} + \beta_6\text{Unemployed}_{ij} + \beta_7\text{Inactive}_{ij} \\ &+ \gamma_1\overline{\text{female}}_j + \gamma_2\overline{\text{age}}_j + \gamma_3\overline{\text{educ}}_j + \gamma_4\overline{\text{partner}}_j + \gamma_5\overline{\text{rural}}_j \\ &+ \delta_1\text{polity}_j + \delta_2\text{GDP_10k}_j + \delta_3\text{incrichest10p}_j + u_j + e_{ij}, \end{aligned}$$

with all variables and terms as previously explained.

Adding the three country characteristics improved model fit relative to the RQ4 contextual model (LRT: $\chi^2(3) = 22.17$, $p < 0.001$). We then assessed whether the contextual country-mean terms remained necessary by comparing the full model with a nested model excluding all contextual terms. The LRT was not significant ($p = 0.27$), indicating that the contextual terms could be removed without a statistically detectable loss of fit. However, as RQ5 requires estimating macro effects *after controlling for individual characteristics and their contextual effects*, we chose to retain the contextual terms in the final model to ensure that country-level coefficients are interpreted net of both compositional and contextual influences.

Interpretation of final model parameters. All interpretations are conditional on the other individual, contextual, and macro-level covariates in the model. Tabular results can be found in Table A6.

- **Female ($\beta_1 = 2.54$).** Women score on average 2.54 points higher than men. This effect remains virtually unchanged from models used in RQ4 and RQ2, indicating that gender differences are not explained by country development or compositional factors.
- **Age ($\beta_2 = -0.082$).** Each additional year of age is associated with a 0.082-point decrease in emancipative values. This effect is stable relative to earlier models.
- **Education ($\beta_3 = 1.25$).** Each additional level of education increases emancipative values by approximately 1.25 points. The magnitude remains strong and largely unchanged after adding macro variables.
- **Partnered ($\beta_4 = -1.20$).** Individuals living with a partner score about 1.20 points lower than those not partnered, holding other factors constant. This association is consistent across specifications.
- **Rural residence ($\beta_5 = -1.61$).** Rural residents score 1.61 points lower than urban residents. This remains stable after macro controls.
- **Employment status.** Relative to employed individuals:
  - Unemployed individuals score 1.08 points lower.
  - Economically inactive individuals score 0.97 points lower.

  These differences persist across model stages.
- **Contextual (country-mean) predictors.** After introducing macro-level variables, none of the country-mean predictors retain statistical significant as seen in RQ4. Their coefficients are small relative to their standard errors, and joint testing indicates they do not improve model fit. Compared to RQ4, this suggests that earlier contextual effects were partly capturing broader macro-level development differences.
- **GDP per capita ($\delta_2 = 3.98$).** A $10,000 increase in GDP per capita is associated with a 3.98-point increase in emancipative values, net of all individual and contextual controls. This is the strongest country-level predictor in the model.
- **Polity ($\delta_1 = 0.16$).** A one-unit increase in the democracy score is associated with a 0.16-point increase in emancipative values, but the effect is small and not statistically distinguishable from zero in this specification.
- **Income inequality ($\delta_3 = 0.34$).** A one-unit increase in the income share of the richest 10% is associated with a 0.34-point increase in emancipative values. This is only marginally significant.
- **Explaining country differences.** The inclusion of country characteristics reduces the country-level variance by approximately 34.3% relative to RQ4. As such, they account for a notable portion of cross-national variation in emancipative values, though residual variance exists.

4 Limitations

4.1 Causal interpretation and endogeneity

The analyses are based on observational cross-sectional survey data, and therefore the estimated coefficients should be interpreted as associations rather than causal effects. Although the multilevel framework separates within- and between-country variation, it does not allow us to determine the direction of causality. For example, while higher education is associated with higher emancipative values, it is also possible that individuals with more liberal value orientations are more likely to pursue further education. This potential reverse relationship limits causal interpretation.

A related concern is that omitted variable bias may arise if unobserved factors are correlated with both the predictors and emancipative values. At the country level, structural features such as welfare

institutions, levels of economic development, or broader cultural traditions may influence both demographic composition and value orientations. If these factors are not fully captured by the included covariates, the estimated contextual effects may be biased.

This concern is particularly relevant for the macro-level predictors included in RQ5. If unobserved cultural or historical factors influence both GDP per capita and emancipative values, the estimated effect of GDP may partially capture these omitted influences. In such cases, the assumption that the country-level random intercept is independent of the included country-level covariates may be violated, leading to biased estimates.

4.2 Data and modelling assumptions

The analysis relies on self-reported survey measures, which may raise concerns about measurement equivalence across countries. Concepts such as "having a partner" or attitudes related to freedom and equality may be interpreted differently across cultural contexts.

The model also assumes linear relationships between continuous predictors and emancipative values, as well as normally distributed random effects. Although these assumptions are standard in multilevel modelling, violations of them may lead to biased estimates or incorrect standard errors.

5 Non-Technical Summary

The purpose of this study was to examine how individual characteristics and national context are associated with emancipative values, which refer to attitudes emphasising freedom of speech, equality, autonomy, and individual choice, rated from 0 to 100. Using multilevel modelling, we were able to distinguish differences between individuals within countries from differences between countries.

We find substantial cross-national variation: around 38% of the total differences in emancipative values arise from differences between countries rather than differences between individuals. This indicates that where people live matters greatly for their values. At the individual level, each additional level of education is associated with roughly a 1-point increase in emancipative values on a 0–100 scale. Age is modestly associated with slightly lower values, while individuals living in rural areas or outside employment report somewhat lower values than their urban or employed counterparts. Women, on average, report higher emancipative values than men.

However, individual characteristics explain less than 10% of the gap between countries. Even after accounting for demographic composition, large differences remain across societies. Once we examine country-level characteristics, economic development is revealed as the main driver. A $30,000 difference in GDP per capita corresponds to nearly a 12-point difference in emancipative values, far larger than the effects associated with political systems or income inequality. Adding macroeconomic indicators reduces cross-national differences by about one-third, though some variation remains unexplained.

An important pattern is that while older individuals tend to report slightly lower emancipative values, societies with higher average ages tend to be more emancipative overall. This suggests that broader modernisation processes outweigh individual life-cycle effects. Gender also shapes how experiences matter, as women gain more from education but experience somewhat stronger disadvantages associated with partnership or unemployment.

Overall, the findings suggest that while personal characteristics matter, broader economic conditions play a particularly important role in shaping national value climates.

A Appendix

Figures

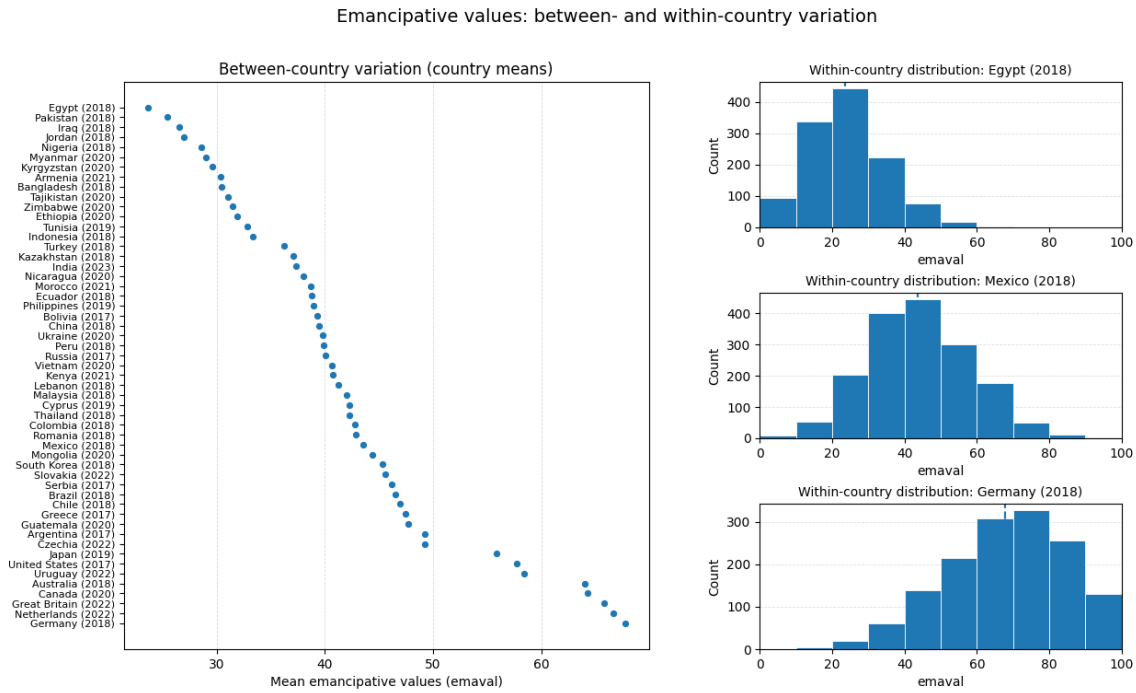

Figure A1: Between and within-country variation in emancipative values (2.2)

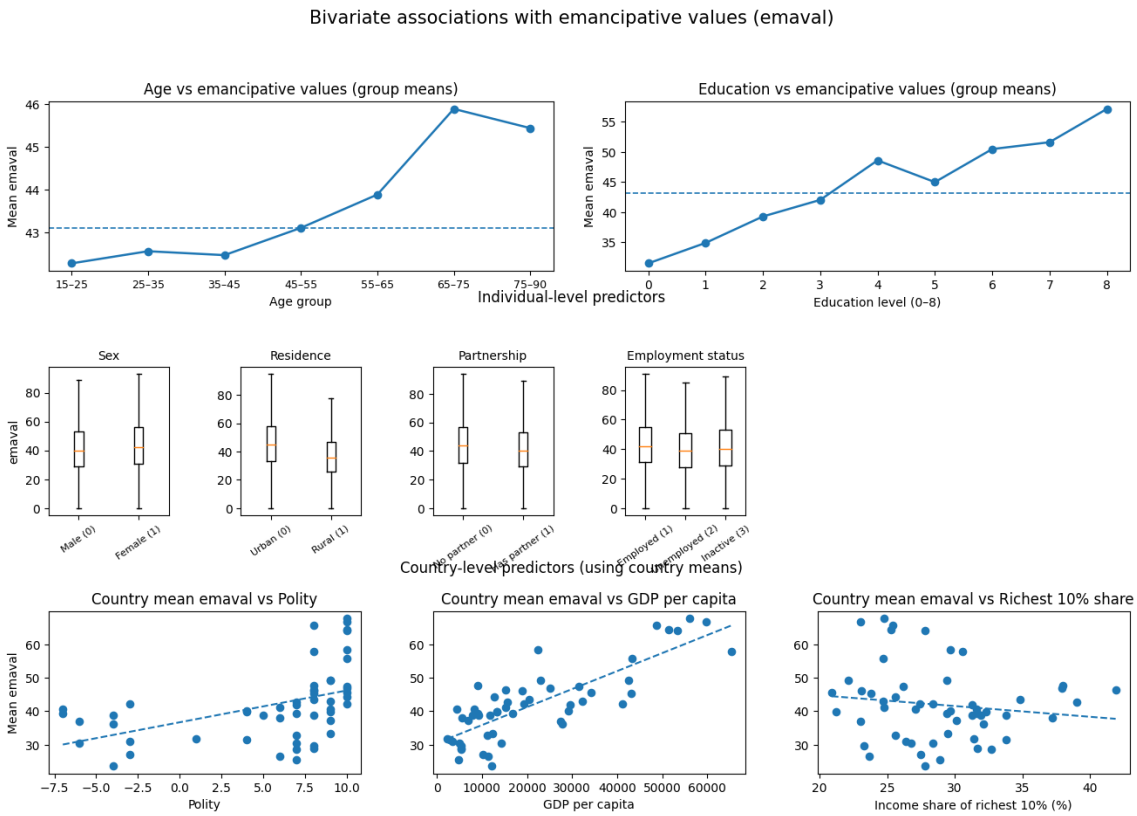

Figure A2: Bivariate associations with emancipative values (2.3)

Tables

Table A1: Random intercept model including individual-level predictors (RQ2)

Predictor	Estimate	Std. Error	p-value
Intercept	41.52	1.44	< 0.001
female	2.54	0.10	< 0.001
age	-0.081	0.003	< 0.001
educ	1.25	0.028	< 0.001
partner	-1.21	0.107	< 0.001
Unemployed	-1.08	0.192	< 0.001
Inactive	-0.97	0.115	< 0.001
rural	-1.61	0.114	< 0.001

Table A2: Likelihood ratio tests for gender interactions (RQ3)

Interaction added	χ^2	df	p
female × educ	68.59	1	< 0.001
female × age	55.84	1	< 0.001
female × rural	8.35	1	0.004
female × partner	26.05	1	< 0.001
female × empstat	34.75	2	< 0.001
All interactions (joint test)	145.37	6	< 0.001

Table A3: Full gender interaction model (RQ3)

Variable	Estimate	Std. Error	p
Intercept	41.059	1.447	< 0.001
female	3.790	0.388	< 0.001
age	-0.068	0.005	< 0.001
educ	1.072	0.038	< 0.001
partner	-0.778	0.167	< 0.001
rural	-1.473	0.158	< 0.001
Unemployed	-0.545	0.262	0.037
Inactive	-0.303	0.190	0.110
female × age	-0.031	0.006	< 0.001
female × educ	0.319	0.051	< 0.001
female × partner	-0.864	0.218	< 0.001
female × rural	-0.268	0.210	0.203
female × Unemployed	-1.144	0.375	0.002
female × Inactive	-0.942	0.240	< 0.001

Table A4: Likelihood ratio tests for individual contextual effects (RQ4)

Contextual term added	χ^2	df	p
age_cm	28.255	1	< 0.001
educ_cm	12.006	1	< 0.001
partner_cm	4.665	1	0.031
rural_cm	23.978	1	< 0.001
female_cm	1.561	1	0.211
All contextual terms (joint test)	48.61	5	< 0.001

Table A5: Full contextual model (RQ4)

Variable	Estimate	Std. Error	p
Intercept	38.921	12.300	0.002
female	2.543	0.101	< 0.001
age	-0.082	0.003	< 0.001
educ	1.253	0.028	< 0.001
partner	-1.203	0.107	< 0.001
rural	-1.605	0.114	< 0.001
female_cm	-58.597	23.178	0.012
age_cm	1.194	0.232	< 0.001
educ_cm	0.806	1.210	0.505
partner_cm	-26.504	11.745	0.024
rural_cm	-9.062	6.279	0.149
Unemployed	-1.083	0.192	< 0.001
Inactive	-0.974	0.115	< 0.001

Table A6: Final model including country characteristics (RQ5)

Variable	Estimate	Std. Error	p
Individual-level predictors			
Intercept	19.092	15.610	0.221
female	2.543	0.101	< 0.001
age	-0.082	0.003	< 0.001
educ	1.253	0.028	< 0.001
partner	-1.203	0.107	< 0.001
rural	-1.605	0.114	< 0.001
Unemployed	-1.081	0.192	< 0.001
Inactive	-0.973	0.115	< 0.001
Contextual (country-mean) predictors			
female_cm	-8.068	21.444	0.707
age_cm	0.445	0.262	0.089
educ_cm	-0.277	1.111	0.803
partner_cm	-15.569	10.382	0.134
rural_cm	-1.740	5.310	0.743
Country-level predictors			
polity	0.156	0.162	0.336
GDP_10k	3.978	0.816	< 0.001
incrichest10p	0.344	0.205	0.093

Statement on Usage of AI

We affirm that each member of this group used generative AI solely for the following purposes:

1. Checking and fixing grammar and spelling errors, including improving sentence structure and paragraph formatting.
2. Debugging R code and formatting our R file to be presentation-ready (most frequently identing and aligning code snippets for ease of readability)
3. LaTeX formatting, including tables, equations, and overall document presentation in Overleaf. All text was initially written in Google Docs before being transferred to LaTeX.

We affirm that no generative AI was used for data analysis, interpretation of results, or any other unauthorised purposes.